

Meeting Summaries of June 29

Date: June 29, 2026

Time: 1:30 - 3:30 Vancouver

Place: TRU, S203

Attendees: Feng, Prof. Jabed, Prof. Yue

Topic: In-person meeting where Feng summarizes his work with the big picture via slides; questioning and feedback from the advisors; discussion on improvements and next work.

Checklist of Accomplishments from Last Meeting

- Feng has emailed TRU Data Science coordinator Dr. Muhammad Tawheed to query the deadline of submitting the master's thesis and finalizing defence date.
- Dr. Muhammad suggested to email TRU graduate office for the official deadline, and Feng emailed the office and there was no response (yet and may not receive one).

Questions, Feedbacks and Suggestions

- The clarity and quality of chemical data needs to be confirmed with Jian and prof. Jan. Feng needs to contact them to check data quality, not only the values but including the units for computation. Then, carefully check the inputted chemical data and the steps of computation in order to duplicate the previous results for comparison.
- In introducing the variable instances, Feng needs to clarify the full name and basic definition of the any feature that audience could ask about. Additionally, it needs to know the basic ecological relation and relationship between these features.
- In data and study case introduction, the flow between the two case studies, corridor case and detroit river case, needs improvement. The DR case objects are a subset from the corridor case, but DR environmental features are superset, one more usable environmental feature, of corridor case.
- There are some issues on slides page-9, the $Var(\cdot)$ formula is written wrong and the notations around matrix are typos.

Action Items

- Feng needs to fix these mentioned issues, bring the fixes and prepare the next proposition for the next meeting.

Summary of Discussion

- Feng gave an overview of objectives of the four computational chapters, introduced the data and study cases background. Answered questions around chemical data quality and especially their units chosen for computation. Feng talked a way of viewing a matrix data structure, an entry as a record pointing to two ticks in row- and column-dimensions. Via this view, Feng explained his understanding about why overlapping or overcounted effects among rawly correlated features exist.

Next Meeting Time

- **Date and Time:** July 2, 2026, 1:30 - 3:30 Vancouver
- **Form:** In-person. Science building, room S203, TRU.